CLASS 10 · CBSE · NCERT MATHEMATICS · DPP SHEET

Chapter 3: Pair of Linear Equations in Two Variables

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Without drawing graphs, state whether the pair $3x + 2y = 5$, $2x - 3y = 7$ is consistent.
2. Find the number of solutions of the pair $x - 2y = 3$, $3x - 6y = 9$.
3. For which value of k does the pair $x + 2y = 3$, $5x + ky = 7$ have no solution?
4. Solve by substitution: $x + y = 17$, $x - y = 5$.
5. Solve by substitution: $2x - y = 4$, $6x - 3y = 12$.
6. Solve by elimination: $2x + 3y = 13$, $5x - 2y = 4$.
7. Solve: $\frac{x}{2} + \frac{y}{3} = 4$, $\frac{x}{4} - \frac{y}{3} = -1$.
8. The sum of two numbers is 48 and one number is five times the other. Find the numbers.
9. 4 pencils and 3 pens cost ₹27, and 3 pencils and 4 pens cost ₹29. Find the cost of one pencil and of one pen.
10. Five years ago a father was seven times as old as his son. Five years from now he will be three times as old. Find their present ages.

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Answer Key

1. $\frac{3}{2} \neq \frac{2}{-3}$, so it is consistent with a unique solution.
2. The ratios are all $\frac{1}{3}$: infinitely many solutions.
3. $\frac{1}{5} = \frac{2}{k}$ gives $k = 10$ (and $\frac{3}{7} \neq \frac{1}{5}$).
4. $x = 11, y = 6$
5. Infinitely many solutions; the equations are the same line.
6. $x = 2, y = 3$
7. $x = 4, y = 6$
8. 8 and 40
9. Pencil ₹3, pen ₹5
10. Father 40 years, son 10 years

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